

Exp No:12

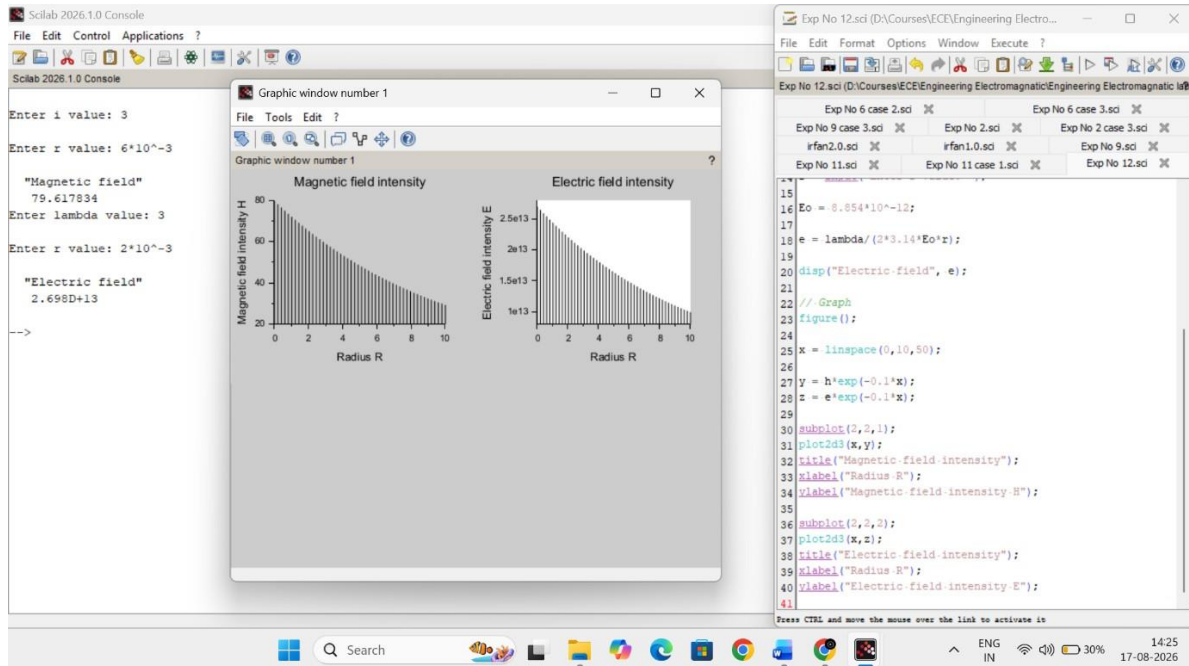
Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

Aim:

To write a program for Magnetic field intensity of infinite long conductor using SCILAB.

Software required:

- SCILAB version 6.0.1



Exr No: 12

magnetic and electric field intensity

$$i = 3 \text{ A}$$

$$\lambda = 3 \text{ m}$$

$$r = 6 \text{ mm}$$

$$z = 2 \text{ mm}$$

magnetic field intensity

$$H = \frac{I}{2\pi r}$$

$$H = \frac{3}{2\pi (6 \times 10^{-3})}$$

$$H = \frac{3}{0.037699}$$

$$H = 79.617 \text{ A/m}$$

Electric field intensity

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$E = \frac{3}{2\pi (8.854 \times 10^{-12})(2 \times 10^{-3})}$$

$$2\pi (8.854 \times 10^{-12})(2 \times 10^{-3}) = 1.127 \times 10^{-3}$$

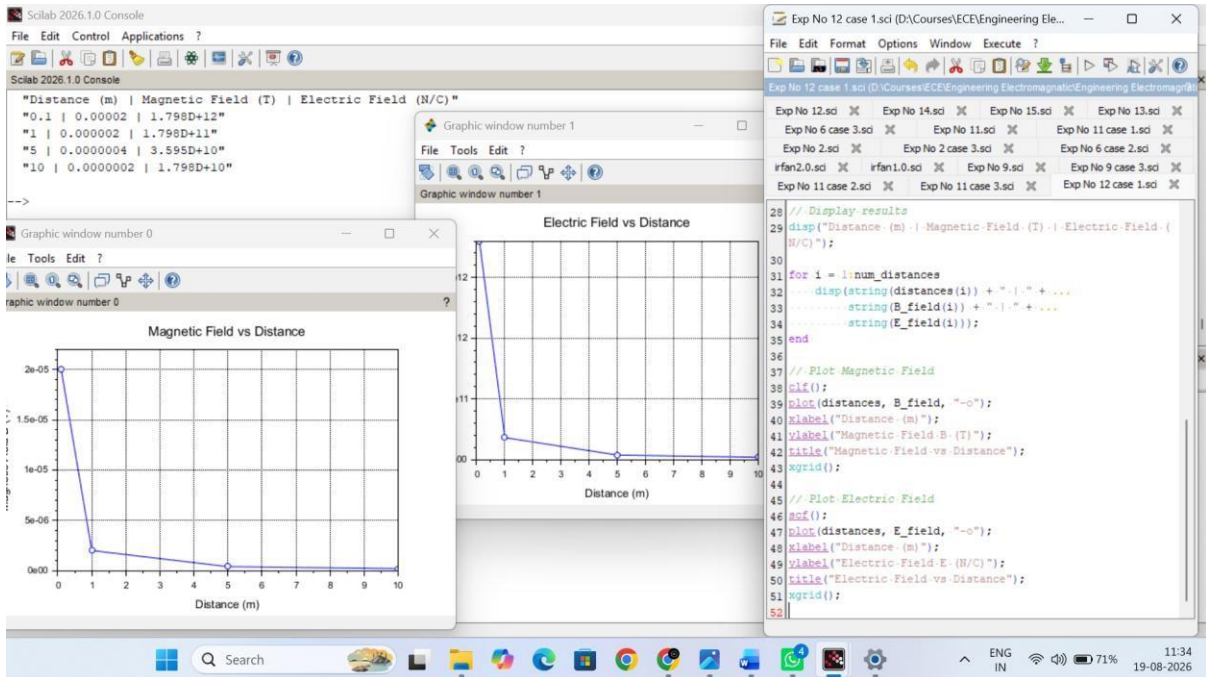
$$E = \frac{3}{1.127 \times 10^{-3}}$$

$$E = 2.698 \times 10^3$$

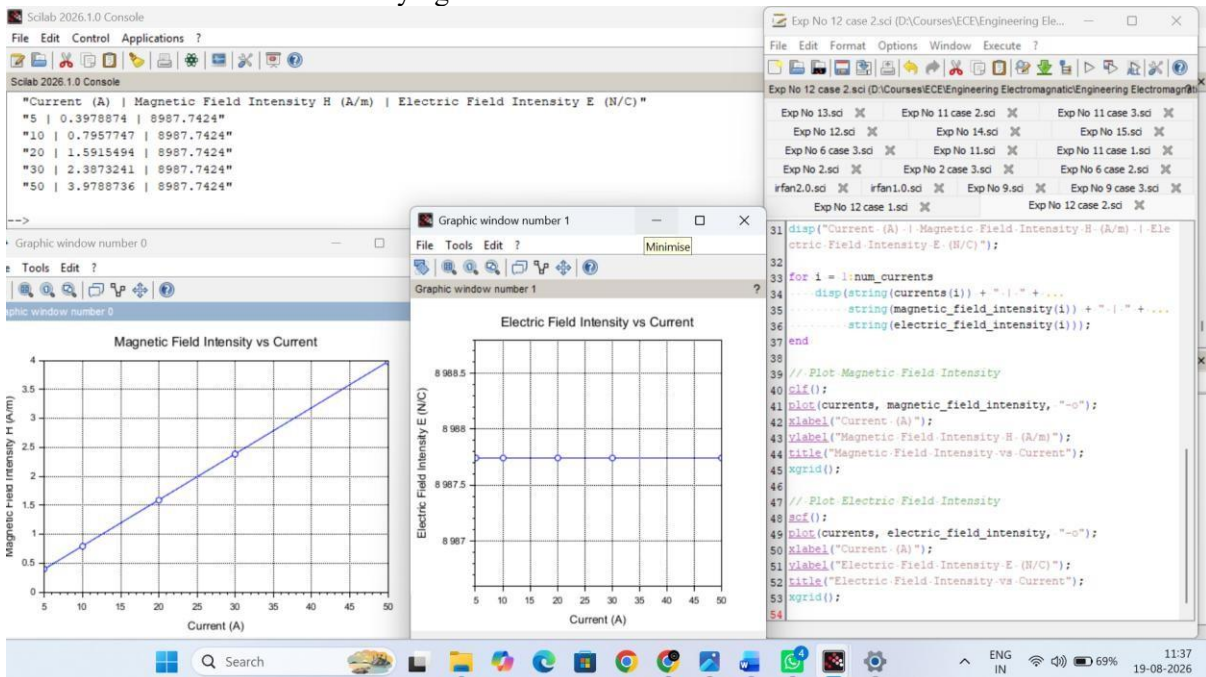
Result:

Thus the program was verified for Magnetic field intensities in ferromagnetic materials using SCILAB was successfully.

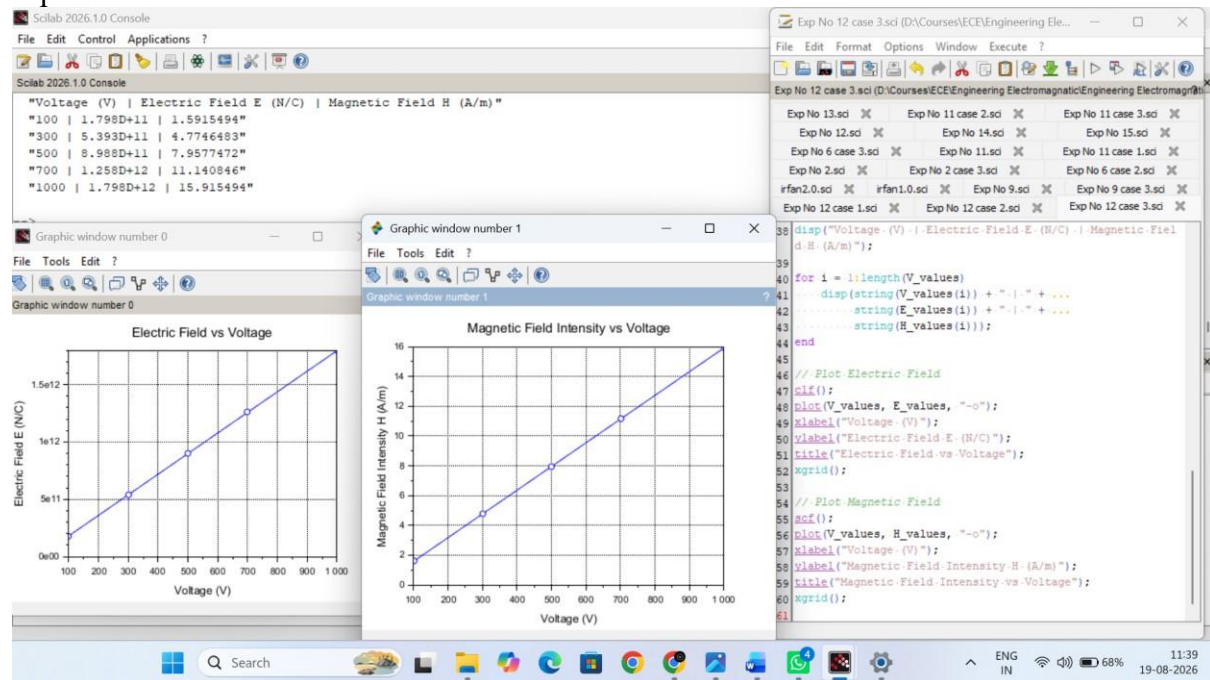
Case 1: To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.



Case 2: To study how the magnetic and electric field intensities change for an infinitely long conductor at a fixed distance H but with varying currents.



Case 3: To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.



Result:

Thus the program was verified for the magnetic and electric field intensities due to infinitely long conductor.

